met_significance_NOSYS

overview · all regions and selections · rank 1 / 169

rank 1 / 169 by endcap / ht40 / D_equal

region	selection	D_equal	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg
endcap	all	0.876	0.896	0.810	0.887	0.859	0.906	0.895	0.939	34,970
endcap	ht40	0.827	0.859	0.746	0.835	0.803	0.871	0.858	0.925	7,390
endcap	bkg split	0.178	below 27,580 / above 7,390							
barrel	all	0.898	0.919	0.831	0.962	0.887	0.936	0.918	0.938	1,089
barrel	ht40	0.833	0.877	0.741	0.926	0.825	0.891	0.881	0.921	220
barrel	bkg split	0.230	below 869 / above 220							
all	all	0.882	0.902	0.814	0.904	0.866	0.913	0.901	0.938	36,059
all	ht40	0.833	0.866	0.749	0.859	0.811	0.879	0.865	0.924	7,610
all	bkg split	0.179	below 28,449 / above 7,610							

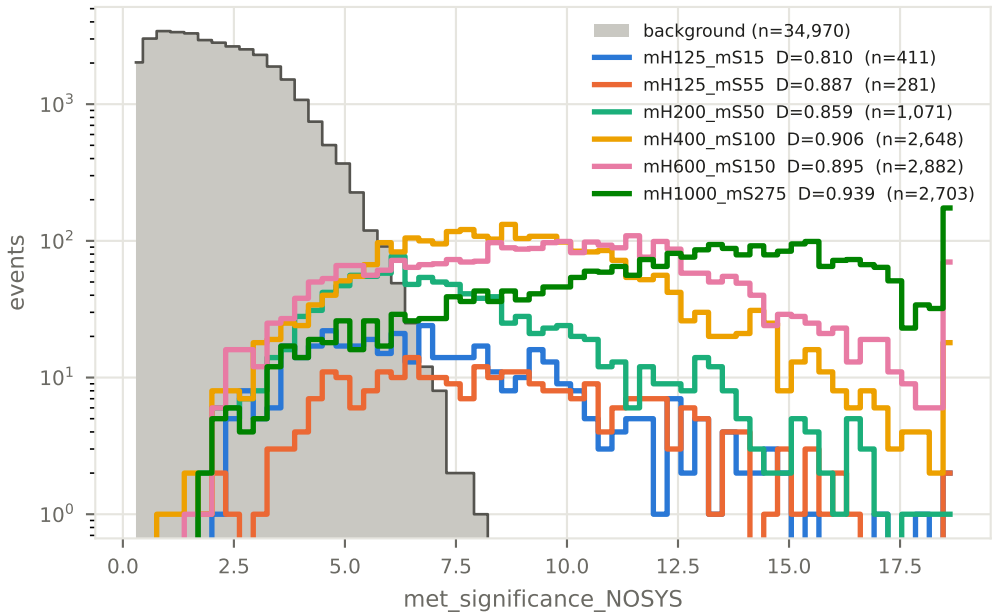
selections: all = every event; ht40 = htmissScalar_NOSYS >= 40 GeV, signal and background
pages: per region -> [all] 2x2, [ht40] 2x2, [background below vs above] 1x2

met_significance_NOSYS

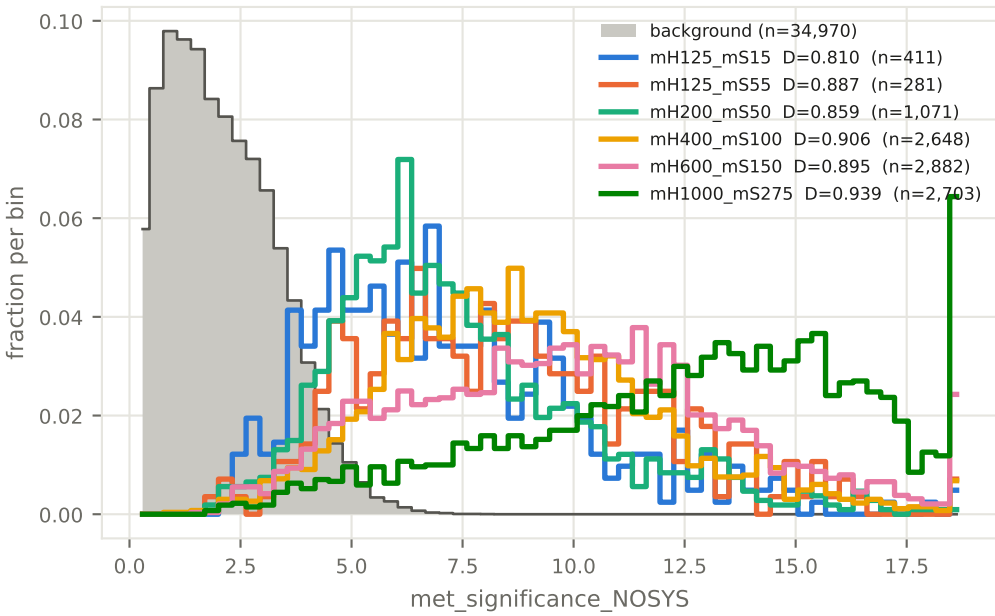
region: ENDCAP · selection: all (every event) · rank 1 / 169

D_equal 0.876 D_pooled 0.896 min 0.810 (mH125_mS15) max 0.939 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

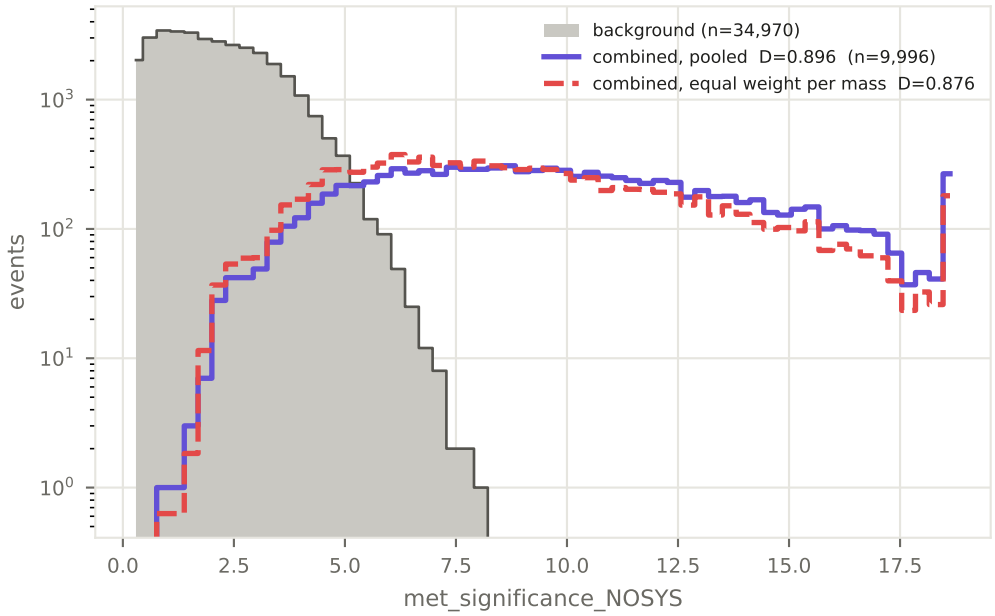
each mass point vs background · raw



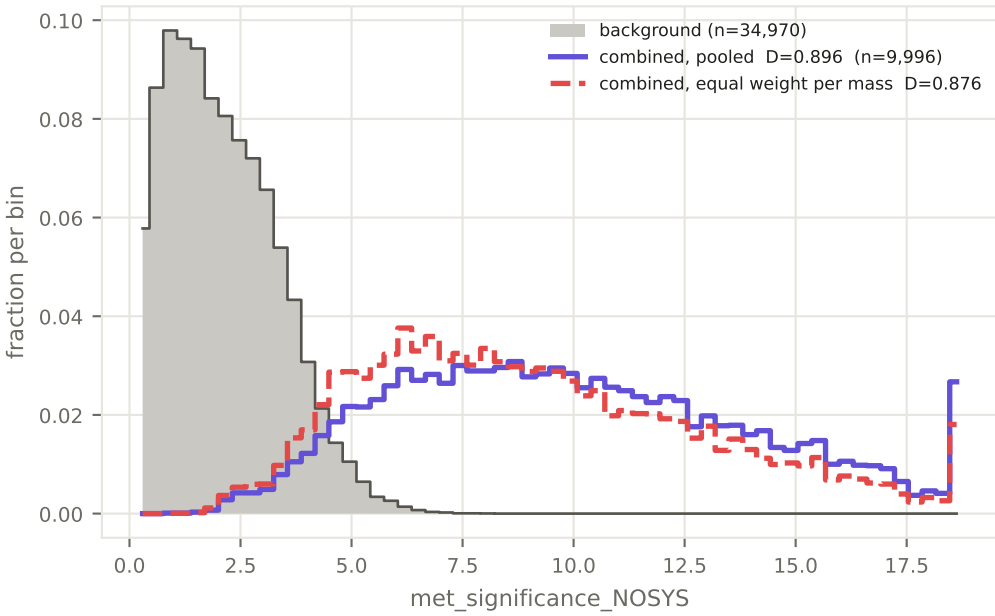
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



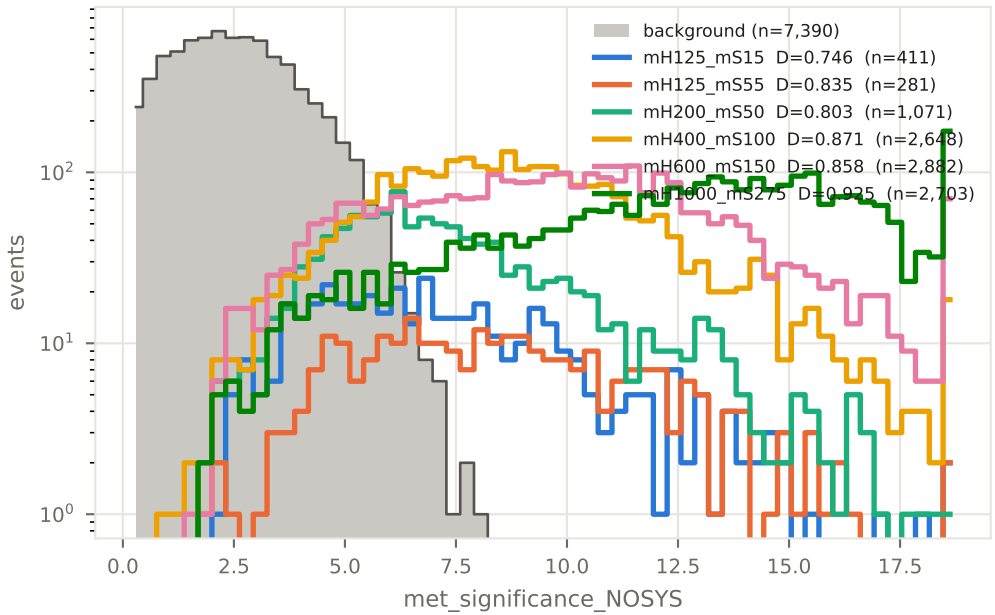
edge bins include under/overflow · NaN and -999 removed

met_significance_NOSYS

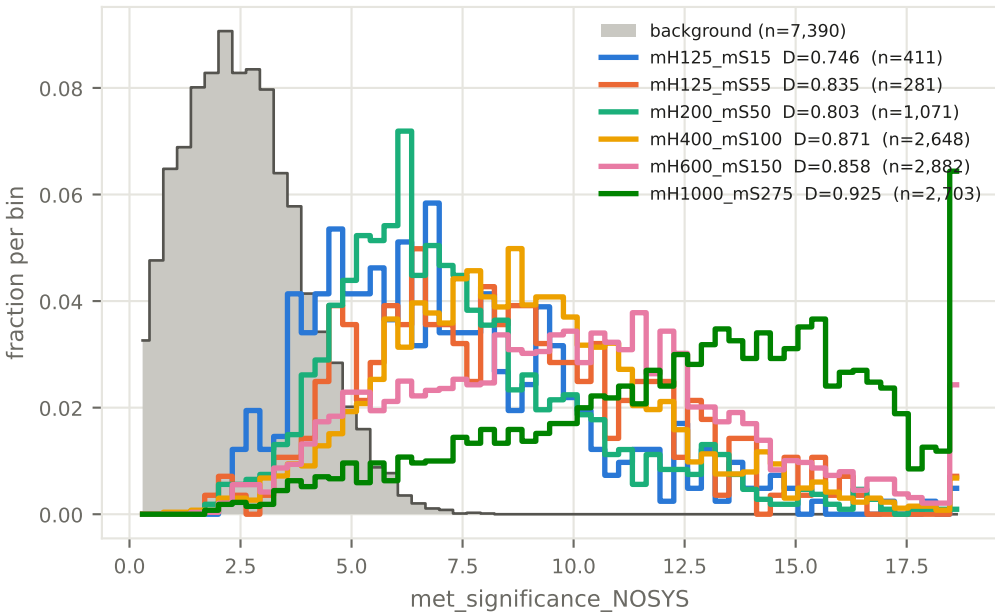
region: ENDCAP · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 1 / 169

D_equal 0.827 D_pooled 0.859 min 0.746 (mH125_mS15) max 0.925 (mH1000_mS275) bkg n = 7,390 sig n = 9,996

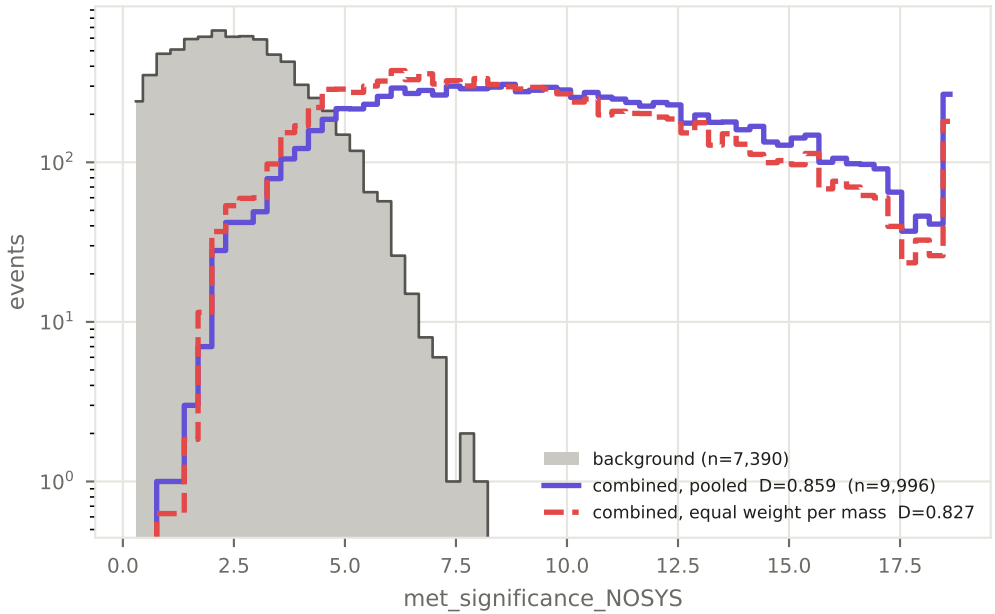
each mass point vs background · raw



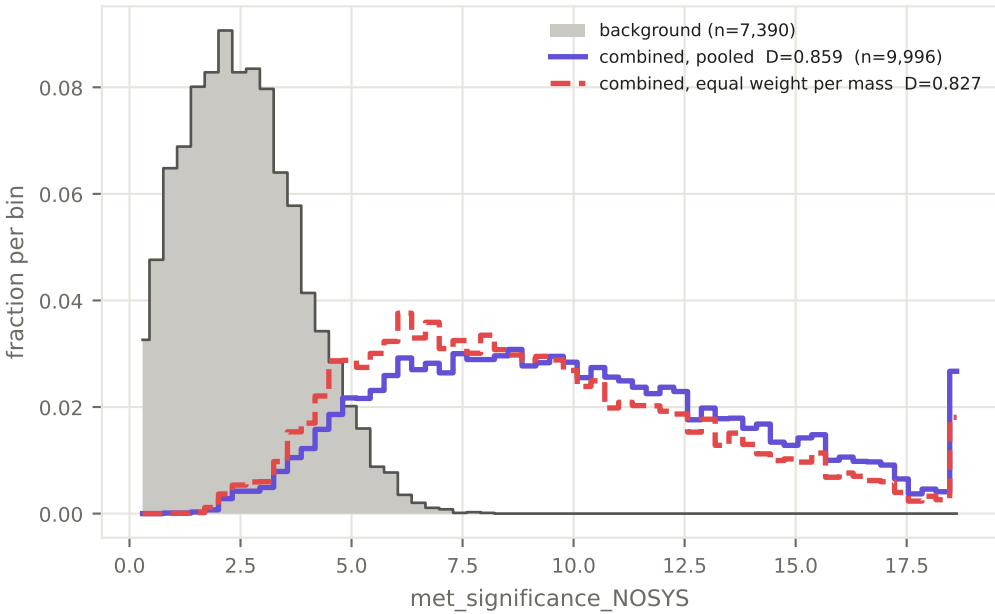
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized

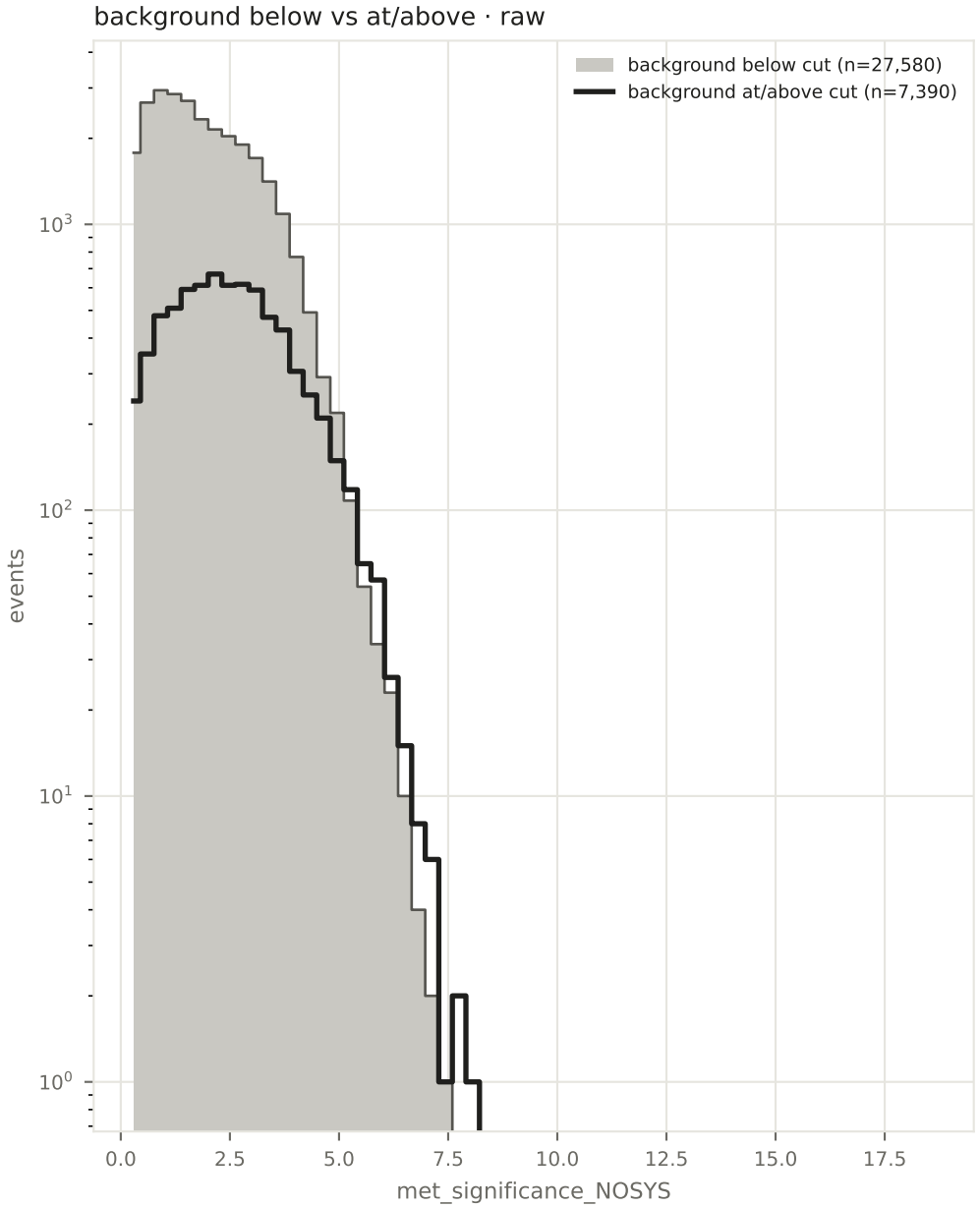


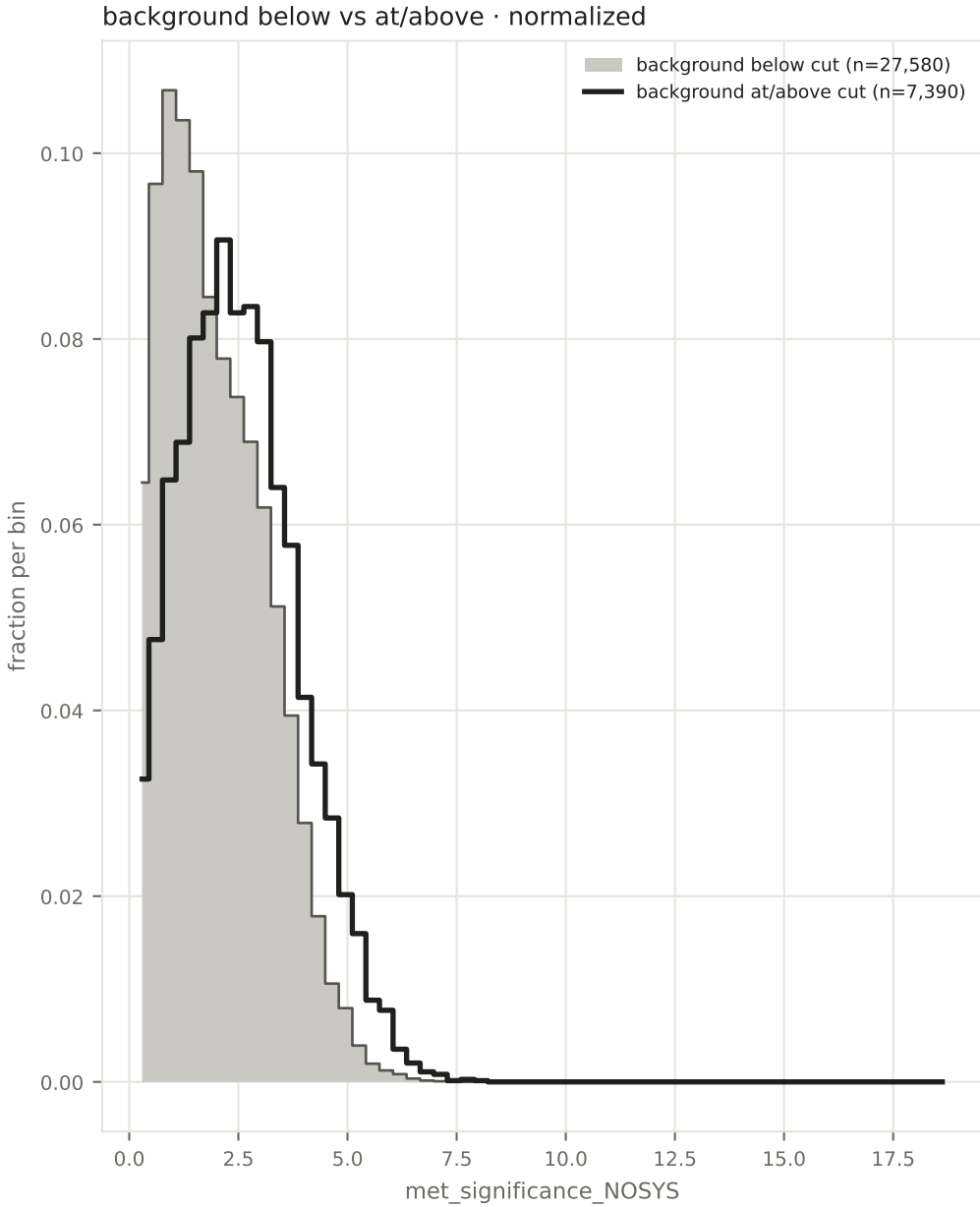
edge bins include under/overflow · NaN and -999 removed

met_significance_NOSYS

region: ENDCAP · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 1 / 169

D(below vs above) 0.178 n below = 27,580 n above = 7,390 large D = background shape changes across the cut



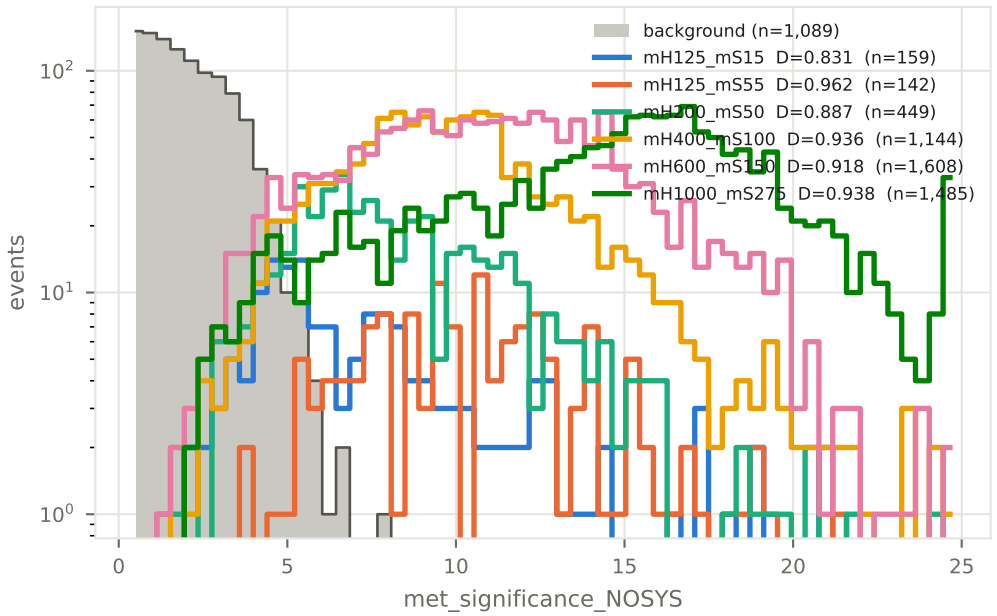


met_significance_NOSYS

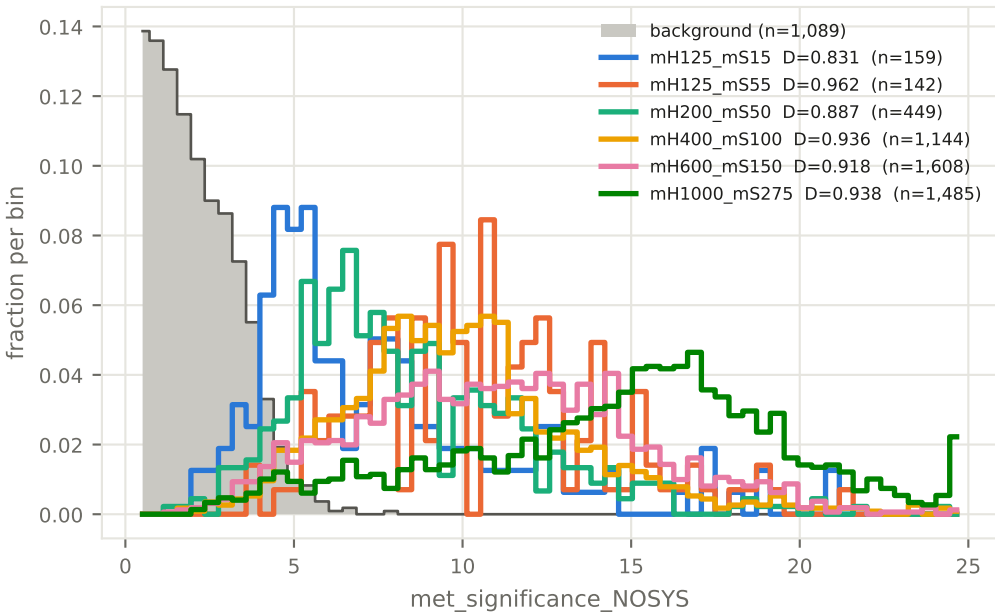
region: BARREL · selection: all (every event) · rank 1 / 169

D_equal 0.898 D_pooled 0.919 min 0.831 (mH125_mS15) max 0.962 (mH125_mS55) bkg n = 1,089 sig n = 4,987

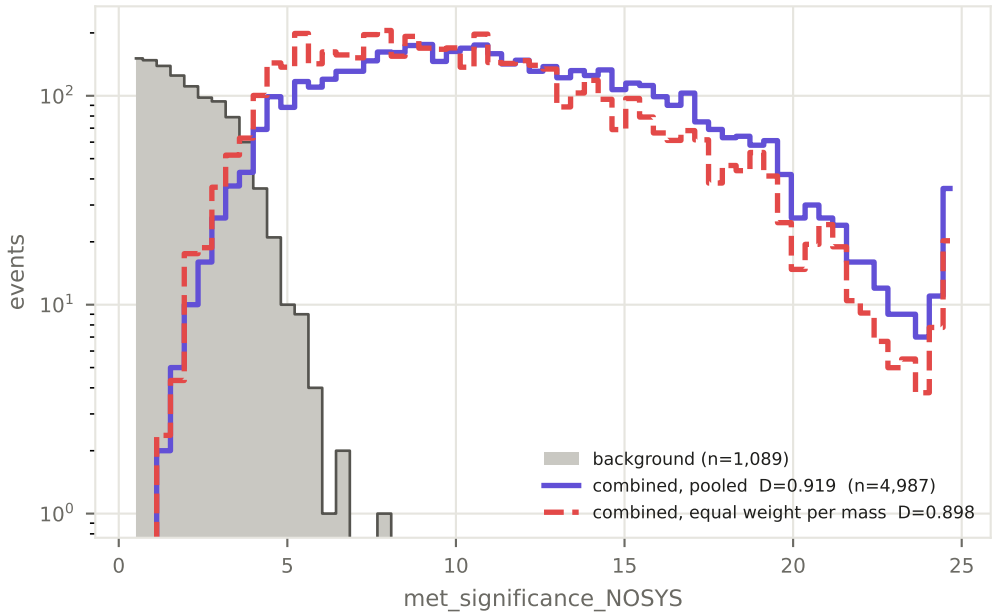
each mass point vs background · raw



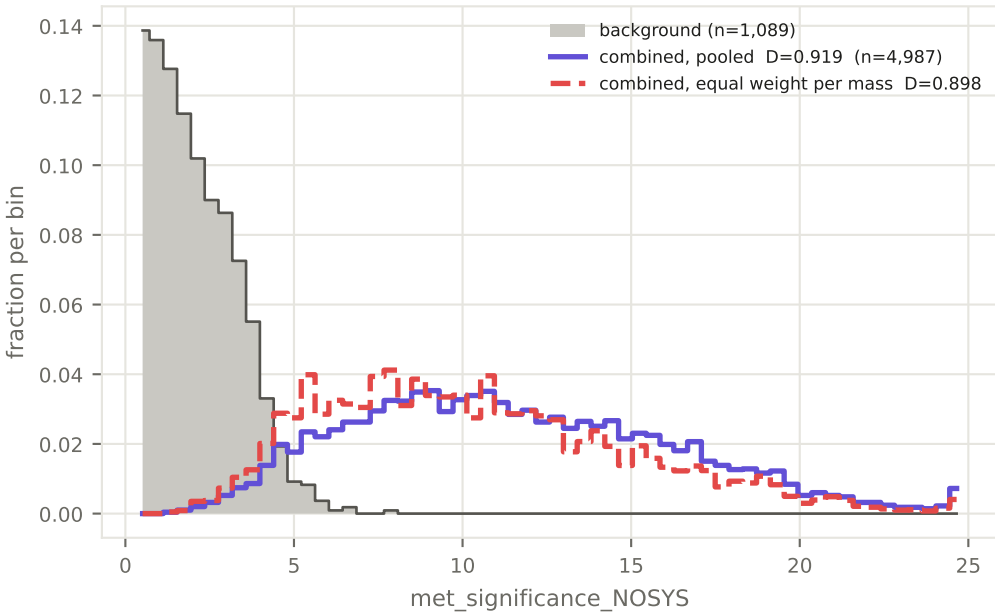
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



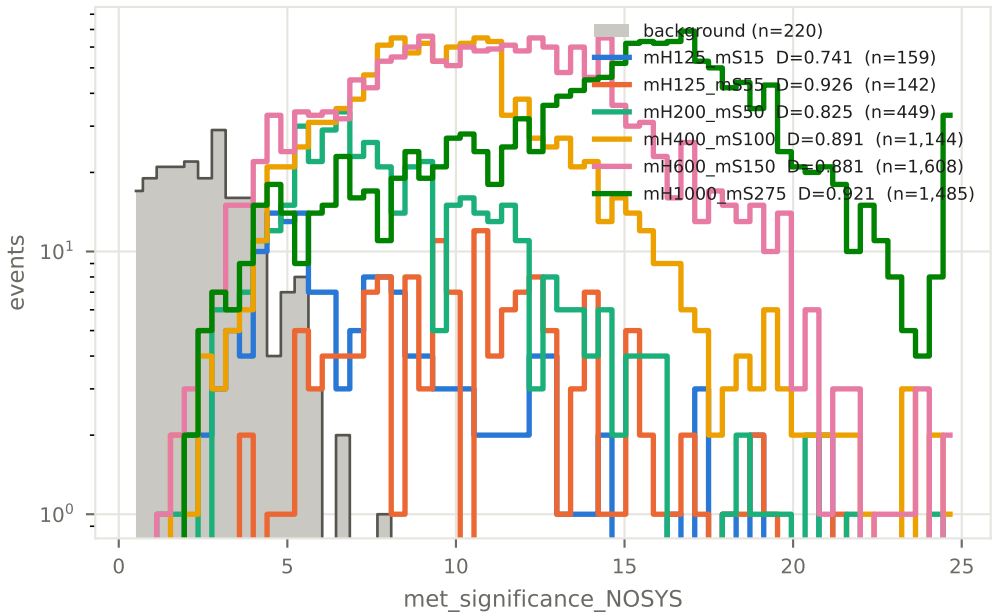
edge bins include under/overflow · NaN and -999 removed

met_significance_NOSYS

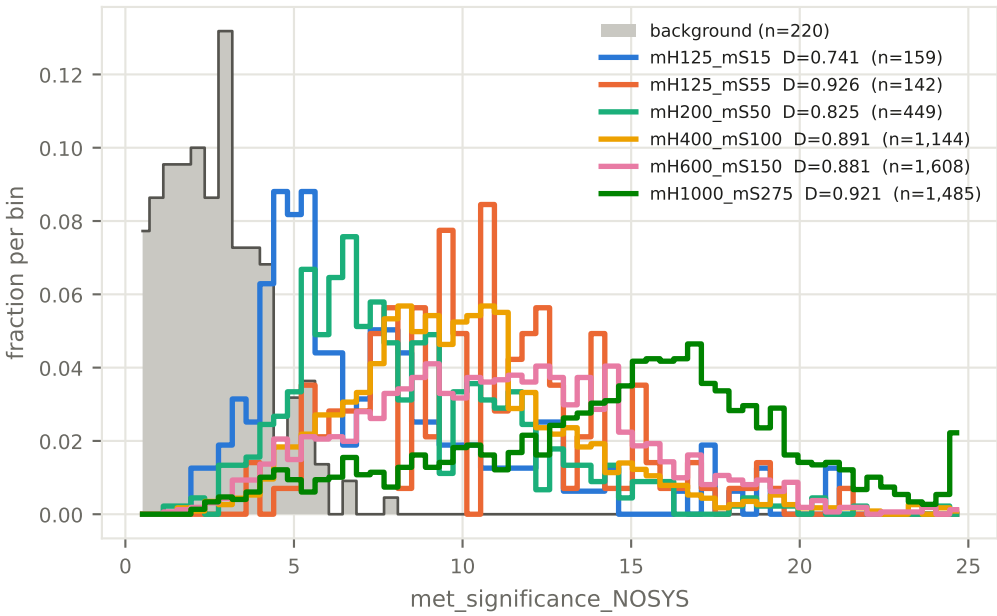
region: BARREL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 1 / 169

D_equal 0.833 D_pooled 0.877 min 0.741 (mH125_mS15) max 0.926 (mH125_mS55) bkg n = 220 sig n = 4,987

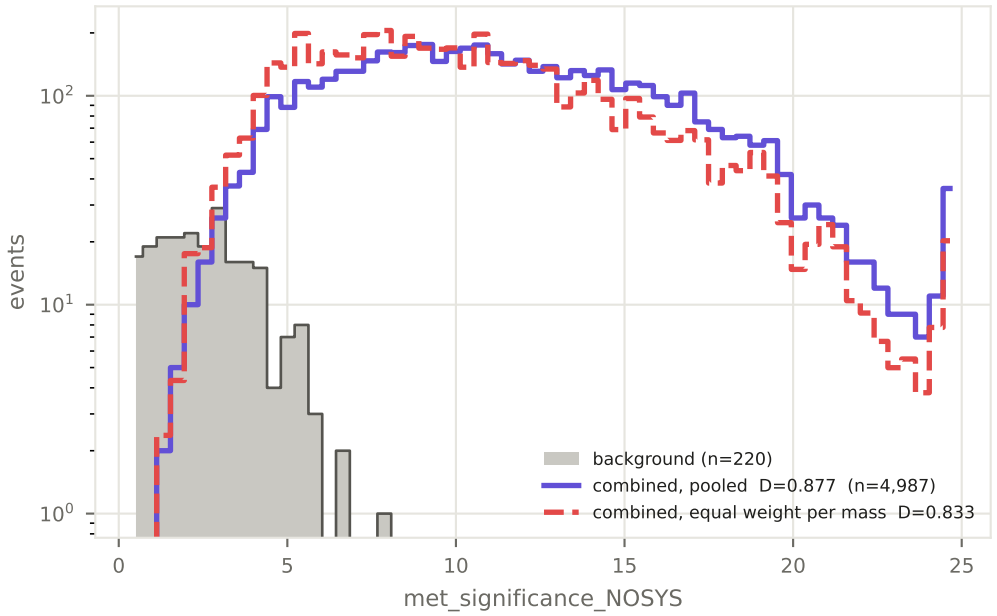
each mass point vs background · raw



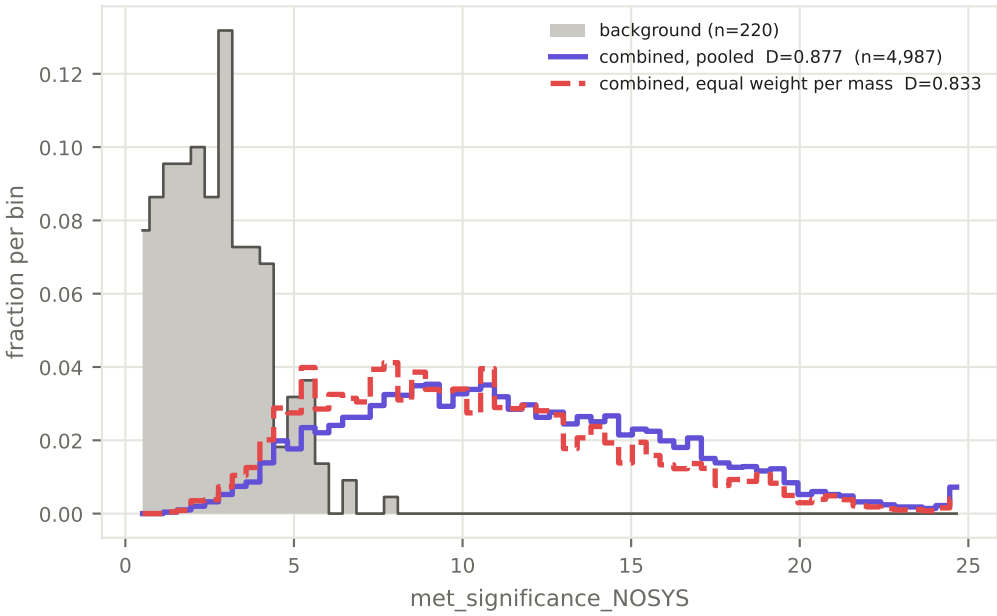
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized

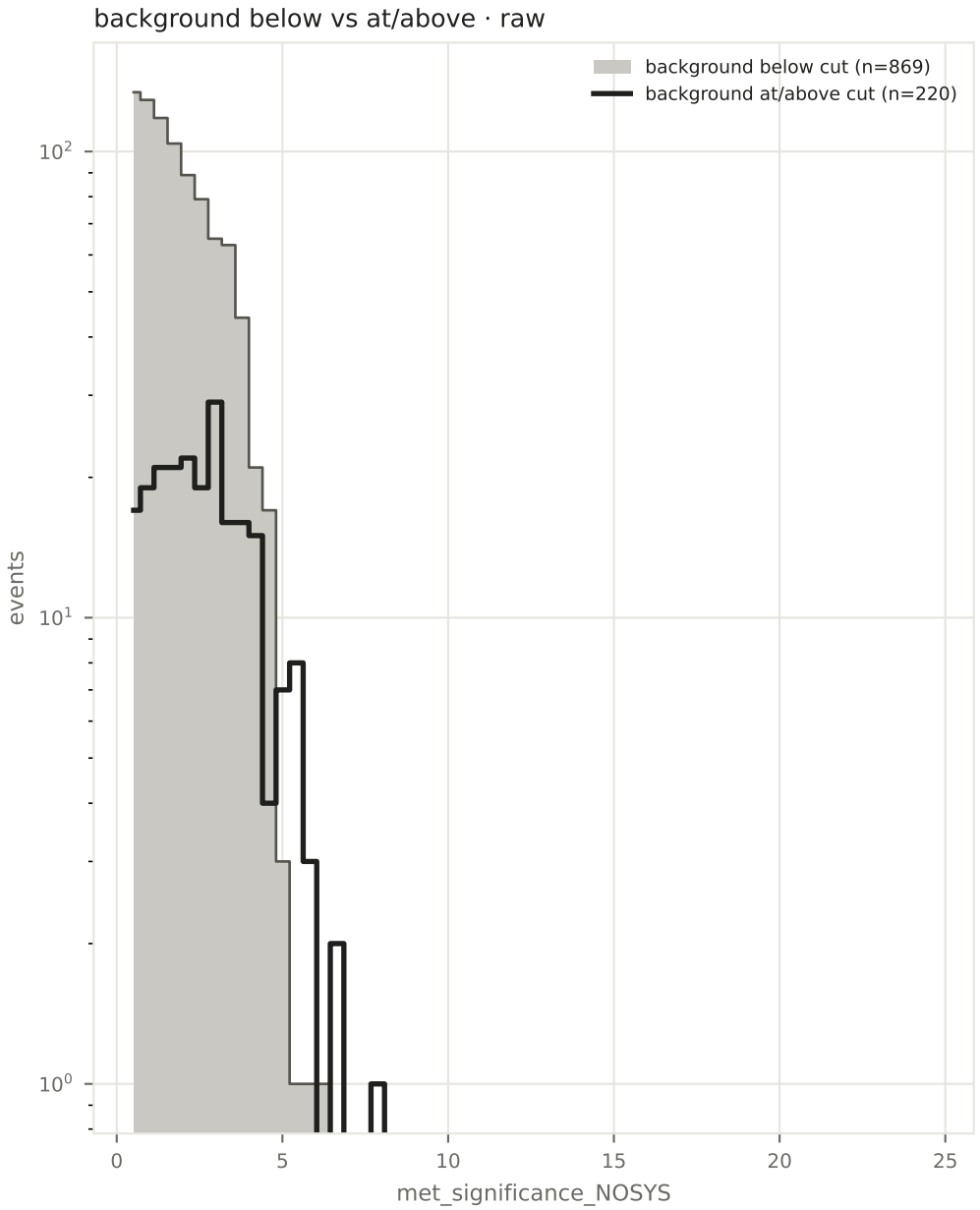


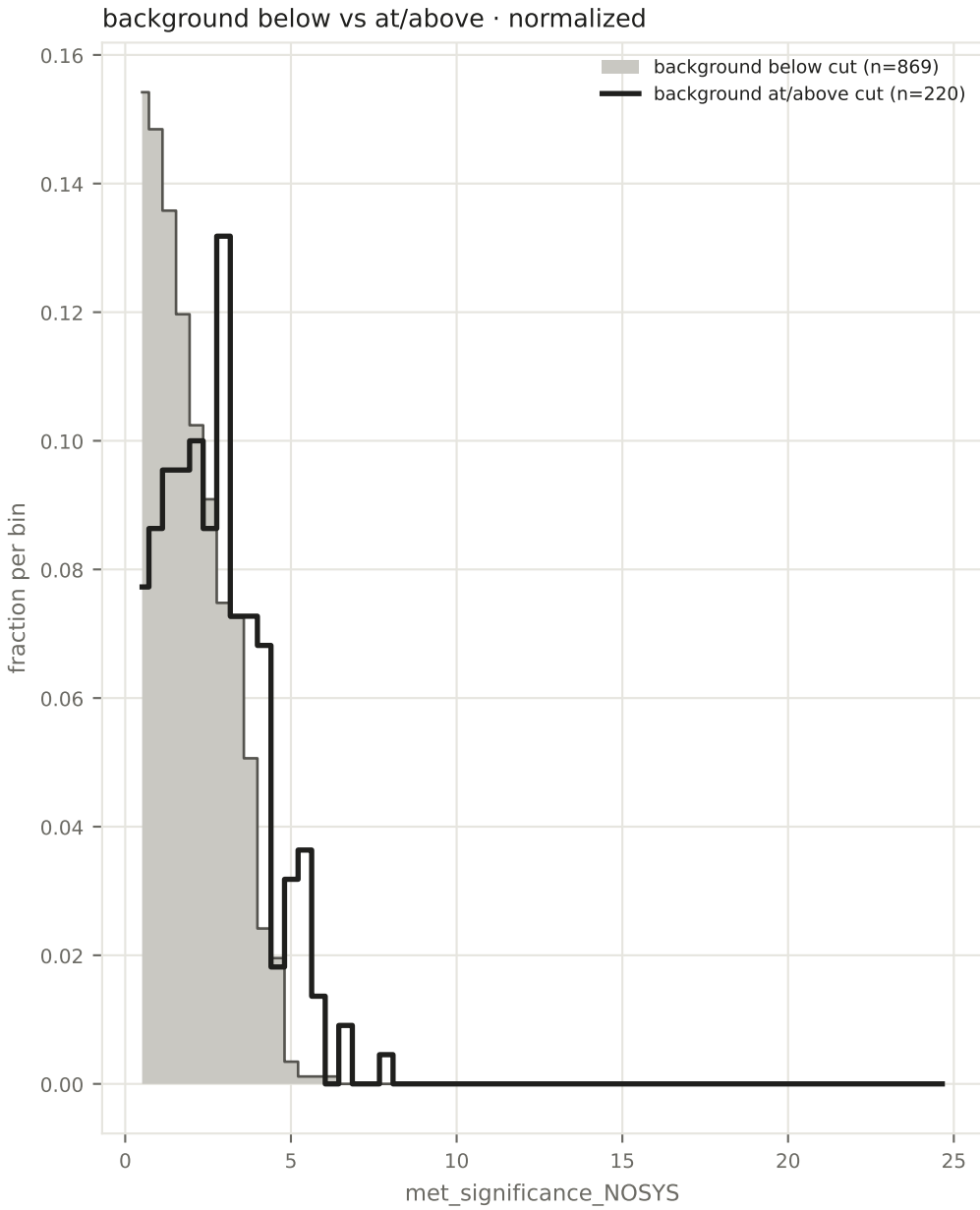
edge bins include under/overflow · NaN and -999 removed

met_significance_NOSYS

region: BARREL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 1 / 169

D(below vs above) 0.230 n below = 869 n above = 220 large D = background shape changes across the cut



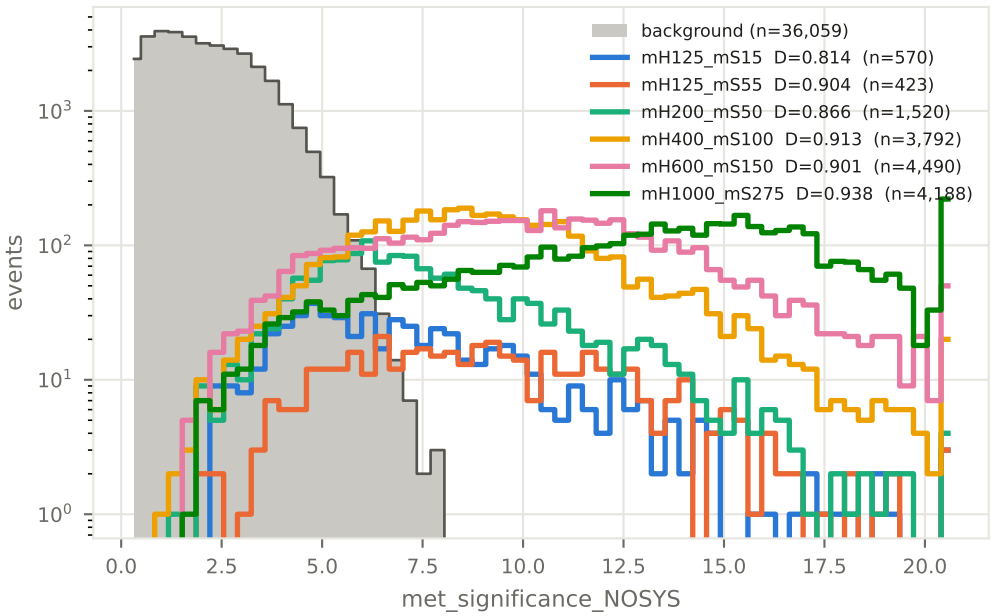


met_significance_NOSYS

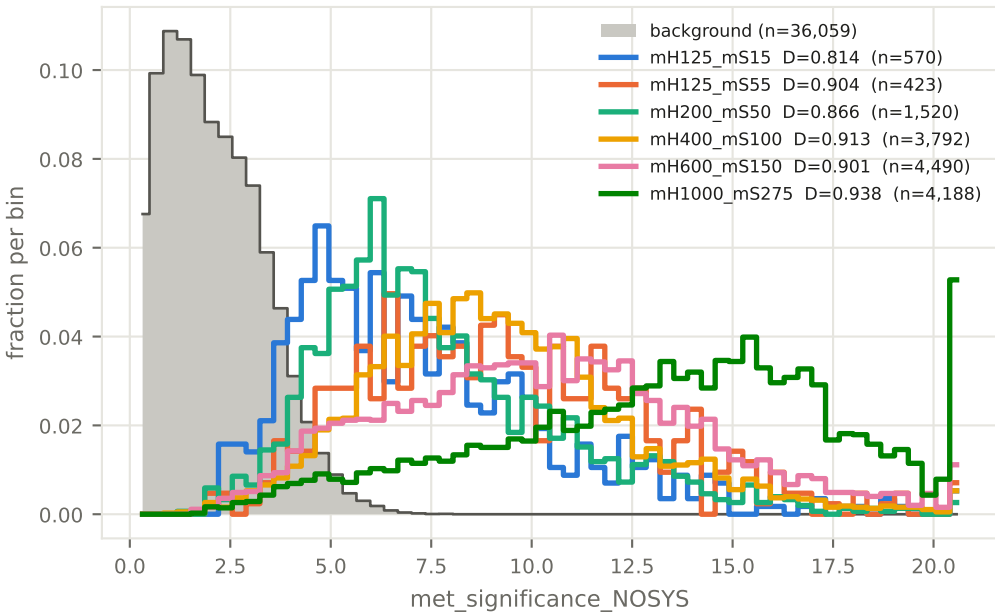
region: ALL · selection: all (every event) · rank 1 / 169

D_equal 0.882 D_pooled 0.902 min 0.814 (mH125_mS15) max 0.938 (mH1000_mS275) bkg n = 36,059 sig n = 14,983

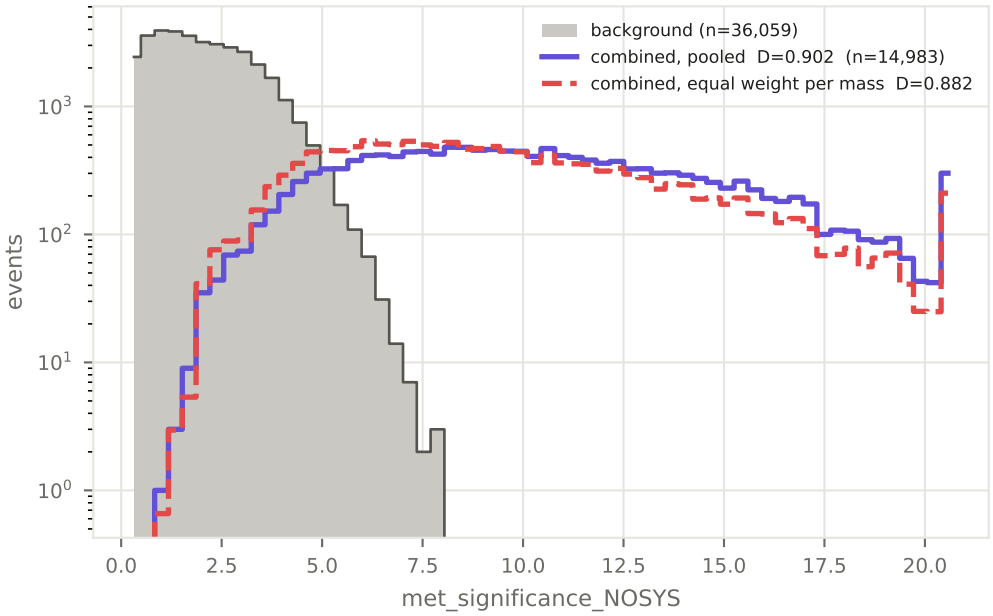
each mass point vs background · raw



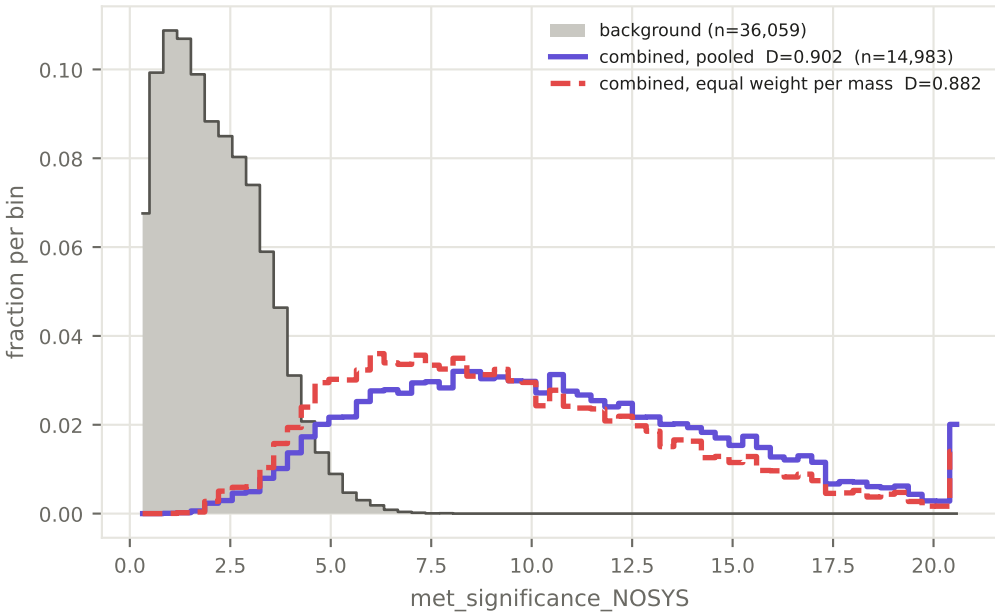
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



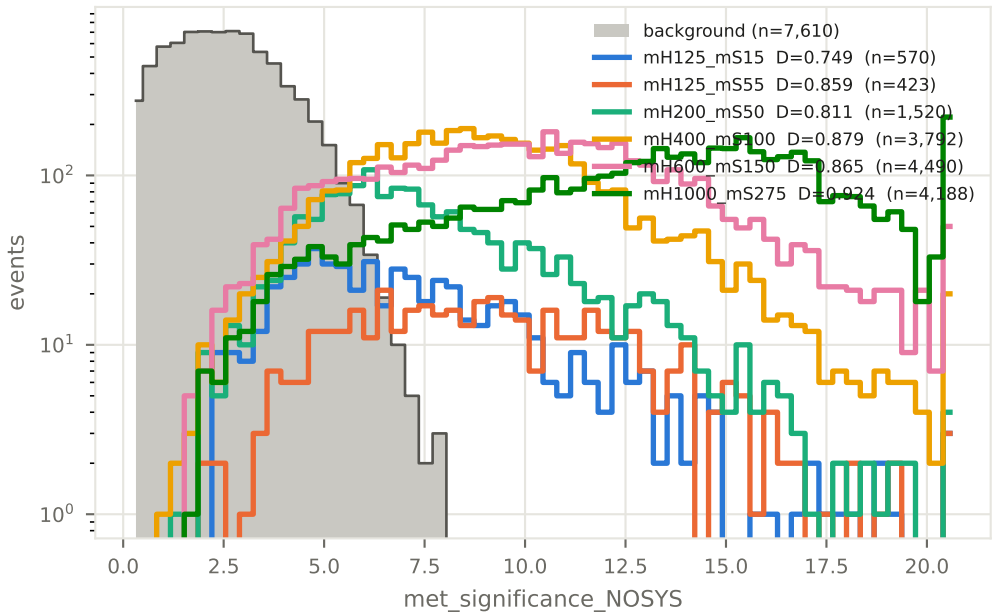
edge bins include under/overflow · NaN and -999 removed

met_significance_NOSYS

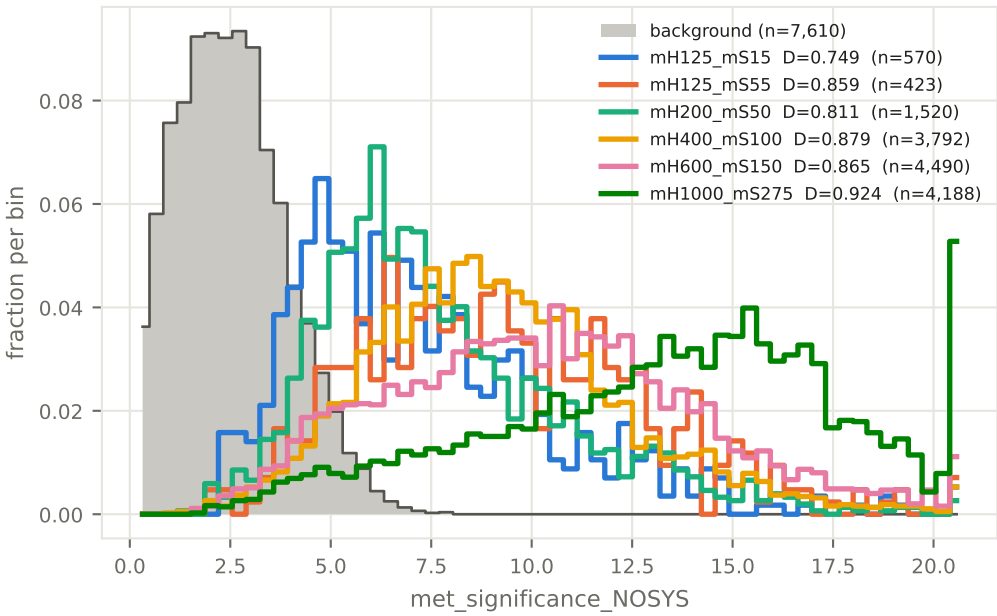
region: ALL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 1 / 169

D_equal 0.833 D_pooled 0.866 min 0.749 (mH125_mS15) max 0.924 (mH1000_mS275) bkg n = 7,610 sig n = 14,983

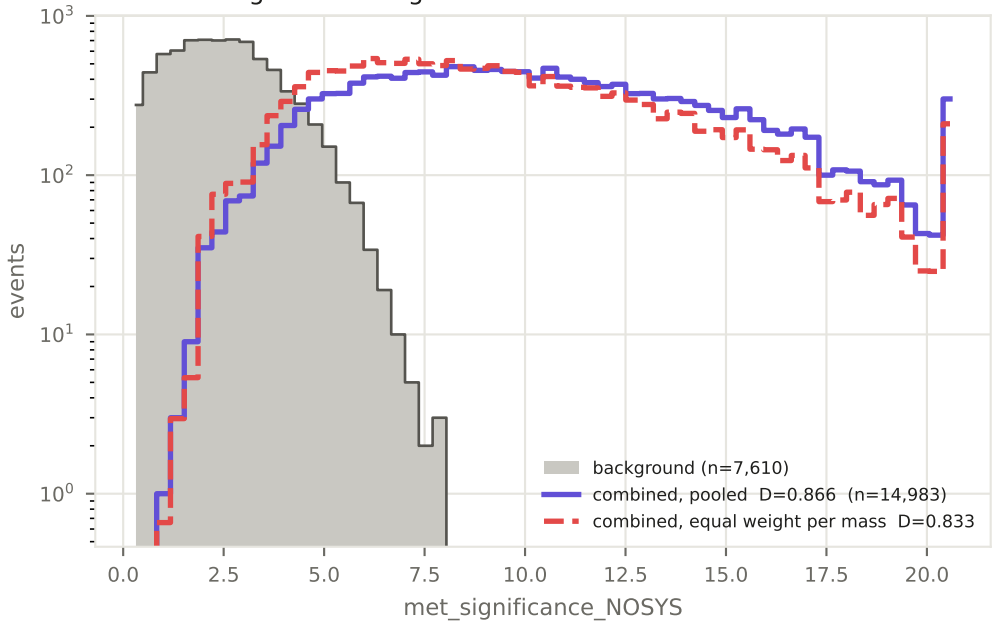
each mass point vs background · raw



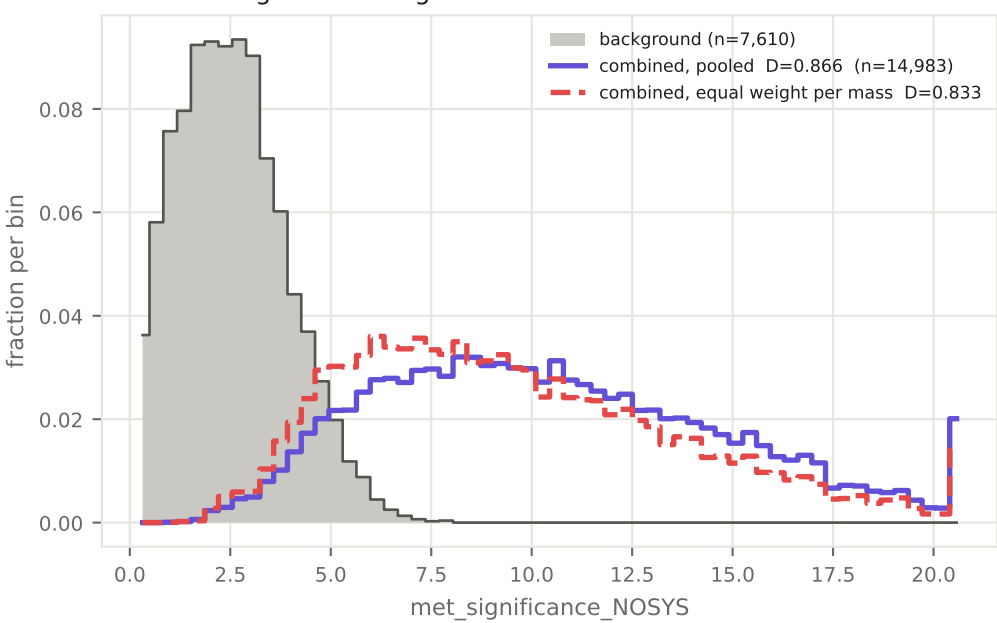
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



edge bins include under/overflow · NaN and -999 removed

met_significance_NOSYS

region: ALL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 1 / 169

D(below vs above) 0.179 n below = 28,449 n above = 7,610 large D = background shape changes across the cut

